→ Given the graph lies between

1 – 4 .

⇒ $P(x) = cx^2$.

integral of $\int P(x) \cdot dx = 1$,

⇒ $\int_{-\infty}^{\infty} cx^2 = \int_{1}^{4} cx^2 = 1$,

⇒ $\frac{cx^3}{3}\Big]_1^4 = 1$,

⇒ $c\left(\frac{63}{3}\right) = 1$

⇒ $c = 1/21$,

⇒ $P(2<x<3)$.

$= \int_2^3 \frac{1}{21} x^2 = \frac{1}{21 \times 3}(27-8)$.

$= \frac{19}{21 \times 3} = \frac{19}{21 \times 3}$.